

Online Server Deployment Manual

Generic Ubuntu VPS installation, HTTPS, backup, update, and recovery

WHO THIS IS FOR	The client's server administrator or contracted technical maintainer
SYSTEM VERSION	1.0
SUPPORT NOTE	Keep this guide beside the clinic host computer or server records.

1. What this package does

This package runs the complete clinic system on a client-chosen Ubuntu server. Caddy provides HTTPS automatically, the frontend and API run in separate containers, and PostgreSQL data remains in a persistent server volume.

STEP 1 Prepare VPS	STEP 2 Point domain	STEP 3 Deploy	STEP 4 Verify
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2. Server requirements

- Ubuntu 22.04 or 24.04 LTS with at least 2 CPU cores, 4 GB RAM, and 30 GB storage.
- A domain name whose DNS A record points to the server's public IP address.
- Inbound ports 22, 80, and 443 allowed by the provider firewall.
- Docker Engine and the Docker Compose plugin installed.
- SSH access using a non-root sudo account and preferably an SSH key.

3. First deployment

1. **Prepare DNS.** Create the domain A record and wait until it resolves to the server
2. **Upload and extract.** Upload the approved online ZIP package, extract it to `/opt/hospital-queue`, and restrict access to the server administrator
3. **Create settings.** In the deploy folder copy `.env.example` to `.env` and replace every `CHANGE_TO` value
4. **Set the domain.** Enter the real `DOMAIN` and `ACME_EMAIL` values. Do not include `http://` or `https://` in `DOMAIN`
5. **Protect the settings.** Run `chmod 600 .env` and `chmod +x clinic.sh`
6. **Install.** Run `sudo ./clinic.sh install` and wait for all services to become healthy
7. **Verify HTTPS.** Open `https://your-domain` and confirm the browser shows a valid secure connection
8. **Sign in.** Use the administrator credentials stored in `.env` and complete the initial doctor and department setup

4. Routine commands

- View status: `sudo ./clinic.sh status`
- Start: `sudo ./clinic.sh start`
- Stop: `sudo ./clinic.sh stop`
- Create backup: `sudo ./clinic.sh backup`
- Install an approved update: `sudo ./clinic.sh update`

5. Backup and off-server storage

1. **Create a database dump.** Run `sudo ./clinic.sh backup`
2. **Confirm the file.** Check that the new SQL file exists and is not empty
3. **Encrypt and copy.** Copy the backup to client-approved encrypted storage outside the VPS
4. **Retain multiple dates.** Keep daily recent backups and at least one older monthly recovery point according to clinic policy
5. **Test restoration.** Use a controlled maintenance window to rehearse restore on a separate test deployment

6. Restore procedure

1. **Declare maintenance.** Prevent staff from changing data during restoration
2. **Take a safety backup.** Back up the current database even if it may contain incorrect data
3. **Restore.** Run `sudo ./clinic.sh restore /secure/path/to/backup.sql`
4. **Restart and verify.** Start the services and verify administrator login, recent appointments, queue history, and reports
5. **Record the incident.** Document the backup used, operator, date, result, and any lost time period

7. Safe updates

- Read the release notes and create an off-server backup first.
- Keep the current `.env` file; never replace it with `.env.example`.
- Upload the new approved application files and run `sudo ./clinic.sh update`.
- Confirm HTTPS, login, booking, queue status, consultation completion, reports, and Socket.IO updates.
- Keep the previous release archive until the update has been accepted.

8. Monitoring and troubleshooting

- Domain does not open: check DNS, provider firewall rules, and `sudo ./clinic.sh status`.
- HTTPS certificate is pending: confirm ports 80 and 443 are public and the domain points to this server.
- Database is unhealthy: do not remove volumes. Check free disk space and container logs before recovery.
- Web page opens but actions fail: check backend health and confirm `ALLOWED_ORIGINS` is generated from the correct DOMAIN.
- Repeated restarts: preserve logs and contact the maintainer; do not repeatedly recreate containers or volumes.

9. Security checklist

- Use SSH keys, disable password-based root login, and apply Ubuntu security updates.
- Never publish .env, SQL backups, or administrator passwords in source control or support messages.
- Allow PostgreSQL only inside the Docker network; do not open port 5432 publicly.
- Limit server access to named maintainers and remove access immediately when responsibilities change.
- Review backups, disk space, certificate renewal, and dependency advisories regularly.

10. Emergency recovery

If the server is lost, create a replacement Ubuntu VPS, point the domain to it, install Docker, deploy the same approved package with secure settings, restore the newest verified off-server backup, and complete the verification checklist before reopening the system to clinic users.